

Cost-Severity Interaction Policy

Synthetic Training Policy - Non-PHI

Defines how billed_amount interacts with diagnosis severity to produce compound risk signals. Covers amount_to_benchmark_ratio, high_cost_flag, cost_overbenchmark_and_highseverity, and mismatch_and_overbenchmark features.

Policy ID	CLAIMOPS-POL-010
Effective Date	2026-04-25
Applies To	Synthetic claims, provider, diagnosis, and cost datasets in this repository
Source Type	Locally generated demo policy; not a real payer coverage policy

Policy Summary

- amount_to_benchmark_ratio compares billed_amount to the procedure-specific expected_cost. A ratio above 1.25 triggers review.
- high_cost_flag is set to 1 when billed_amount exceeds 2x the procedure expected_cost.
- cost_overbenchmark_and_highseverity combines over-benchmark charges with a high-severity diagnosis — the strongest cost-based risk signal.
- mismatch_and_overbenchmark combines specialty-diagnosis mismatch with over-benchmark charges — dual misalignment is a high-confidence denial predictor.

Rules

Rule ID	Requirement
COST-1	PROC1 benchmark expected_cost is 5,000 (Delhi reference pricing). billed_amount above 6,250 (25% over benchmark) triggers OVER_BENCHMARK_REVIEW. billed_amount above 10,000 (2x benchmark) sets high_cost_flag = 1.
COST-2	PROC2 benchmark expected_cost is 15,000 (Mumbai reference pricing). billed_amount above 18,750 (25% over benchmark) triggers OVER_BENCHMARK_REVIEW. billed_amount above 30,000 (2x benchmark) sets high_cost_flag = 1.
COST-3	PROC3 benchmark expected_cost is 9,000 (Bangalore reference pricing). billed_amount above 11,250 (25% over benchmark) triggers OVER_BENCHMARK_REVIEW. billed_amount above 18,000 (2x benchmark) sets high_cost_flag = 1.
COST-4	PROC4 benchmark expected_cost is 8,000 (Hyderabad reference pricing). billed_amount above 10,000 (25% over benchmark) triggers OVER_BENCHMARK_REVIEW. billed_amount above 16,000 (2x benchmark) sets high_cost_flag = 1.
COST-5	PROC5 benchmark expected_cost is 25,000 (Chennai reference pricing). billed_amount above 31,250 (25% over benchmark) triggers OVER_BENCHMARK_REVIEW. billed_amount above 50,000 (2x benchmark) sets high_cost_flag = 1.
COST-6	PROC6 benchmark expected_cost is 1,200 (Ahmedabad reference pricing). billed_amount above 1,500 (25% over benchmark) triggers OVER_BENCHMARK_REVIEW. billed_amount above 2,400 (2x benchmark) sets high_cost_flag = 1.

Rule ID	Requirement
COST-INT-07	high_cost_flag = 1 when billed_amount exceeds 2x the procedure expected_cost. High-cost claims undergo mandatory secondary review regardless of other risk signals.
COST-INT-08	cost_overbenchmark_and_highseverity = 1 when BOTH amount_to_benchmark_ratio > 1.25 AND diagnosis_severity_encoded indicates High severity. This intersection triggers the most intensive manual-review protocol because high-cost, high-severity claims carry the greatest financial and clinical risk.
COST-INT-09	mismatch_and_overbenchmark = 1 when BOTH specialty_diagnosis_mismatch = 1 AND amount_to_benchmark_ratio > 1.25. Dual misalignment — clinical AND financial — is the strongest single denial predictor in the model. These claims require both clinical documentation AND cost justification.

Remediation Guidance

- For over-benchmark claims, verify the billed_amount against the charge master and contract rates.
- For cost-overbenchmark-and-highseverity claims, require both clinical notes AND a cost outlier explanation.
- For mismatch-and-overbenchmark claims, escalate to medical director review before any payer submission.
- Do not include billed_amount or expected_cost in LLM prompts. Use over benchmark or within benchmark descriptors.

RAG Safety Note

This document contains only synthetic policy text. It intentionally avoids patient names, patient identifiers, service dates, and real billed amounts. Use policy_id, rule_id, denial reason, diagnosis category, and procedure status as retrieval metadata.